

9 Comparing Bayesian and Frequentist Analysis

The most widely used elements of “traditional” frequentist inference are confidence intervals and hypothesis tests (a.k.a, null hypothesis significance tests). The numerical results of Bayesian and frequentist analysis are often similar. However, the interpretations are very different.

Example 9.1. Recall Example 7.4 which concerned θ , the population proportion of American adults who have read a book in the last year. Recall the actual study data in which 75% of the 1502 American adults surveyed said they read a book in the last year.

We’ll compare our Bayesian analysis in Example 7.4 to a frequentist analysis.

1. Compute a 98% frequentist confidence interval for θ .
2. Write a clearly worded sentence reporting the confidence interval in context.
3. Explain what “98% confidence” means.
4. Compare the *numerical results* of the Bayesian and frequentist analysis. Are they similar or different?

5. How does the *interpretation* of these results differ between the two approaches?
6. From a frequentist perspective, which value, 0.73 or 0.75, is more plausible for θ , rounded to two decimal places? Explain.
7. From a Bayesian perspective, which value, 0.73 or 0.75, is more plausible for θ , rounded to two decimal places? Explain.

Example 9.2. Continuing Example 9.1. Have more than 70% of Americans read a book in the last year? We'll now compare a Bayesian analysis to a frequentist (null) hypothesis (significance) test.

Recall the actual study data in which 75% of the 1502 American adults surveyed said they read a book in the last year.

1. Conduct an appropriate hypothesis test.
2. Write a clearly worded sentence reporting the conclusion of the hypothesis test in context.

3. Write a clearly worded sentence interpreting the p-value in context.

4. Now back to the Bayesian analysis of Example 7.4. Compute the posterior probability that θ is less than or equal to 0.70.

5. Compare the *numerical values* of the posterior probability and the p-value. Are they similar or different?

6. How does the *interpretation* of these results differ between the two approaches?

In a Bayesian approach

- Parameters are random variables and have distributions.
- Observed data are treated as fixed, not random.
- All inference is based on the posterior distribution of parameters which quantifies our uncertainty in the parameters, given the observed sample data.
- The posterior (or prior) distribution can be used to make probability statements about parameters.
- For example, “95% credible” quantifies our assessment that the parameter is 19 times more likely/plausible to lie inside the credible interval than outside. (Roughly, we’d be willing to bet at 19-to-1 odds on whether θ lies inside the interval.)

- The posterior distribution quantifies a ranking of which values are more or less plausible as a possible value of the parameter, and how much more/less plausible.
- Testing a null hypothesis versus an alternative hypotheses plays a much smaller role

In a frequentist approach

- Parameters are treated as fixed (not random), but unknown numbers
- So it doesn't make sense to make probability statements about parameters
- Data are treated as random
- All inference is based on the sampling distribution of the data which quantifies how the data would behave over many hypothetical samples.
- For example, "95% confidence" is confidence in the procedure: confidence intervals vary from sample-to-sample; over many samples 95% of confidence intervals contain the parameter being estimated.
- Any value inside the confidence interval is a plausible value for the parameter. Any value outside the confidence interval is not a plausible value of the parameter. A value is either plausible or not; the cut-off is arbitrary (and conventionally 95%). Aside from this, there is no concept of some values being more plausible than others.
- Traditional tests of null versus alternative hypotheses don't really answer interesting questions (e.g., they focus on "is there a difference?" as opposed to "is the difference large or meaningful?")
- p-values are confusing

Example 9.3. Recall Example 8.4 in which we assumed body temperatures (degrees Fahrenheit) of healthy adults follow a Normal distribution with unknown mean θ and known standard deviation $\sigma = 1$, and our goal was to estimate θ , the population mean healthy human body temperature.

We performed a Bayesian analysis based on a sample of 208 healthy adults with a sample mean body temperature of 97.7 degrees F.

1. Compute a 98% frequentist confidence interval for θ .
2. Write a clearly worded sentence reporting the confidence interval in context.

3. Compare the *numerical results* of the Bayesian and frequentist analysis. Are they similar or different?
4. How does the *interpretation* of these results differ between the two approaches?
5. From a frequentist perspective, which value, 97.6 or 97.7, is more plausible for θ , rounded to one decimal place? Explain.
6. From a Bayesian perspective, which value, 97.6 or 97.7, is more plausible for θ , rounded to one decimal place? Explain.

Example 9.4. Continuing Example 9.3. Is population mean healthy human body temperature less than 98.6 degrees Fahrenheit? We'll now compare a Bayesian analysis to a frequentist (null) hypothesis (significance) test.

1. Conduct an appropriate hypothesis test.
2. Write a clearly worded sentence reporting the conclusion of the hypothesis test in context.

3. Write a clearly worded sentence interpreting the p-value in context.
4. Now back to the Bayesian analysis of Example 8.4. Compute the posterior probability that θ is greater than or equal to 98.6.
5. Compare the *numerical values* of the posterior probability and the p-value. Are they similar or different?
6. How does the *interpretation* of these results differ between the two approaches?